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Per ***Executive Order 14434***, federal agencies are now using 'Super Intelligence' (SI) in place of 'Artificial Intelligence' (AI). You may see both terms across NIH content during this transition.

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Diabetes - insulin therapy

Insulin is a hormone produced by the pancreas to help the body use and store glucose (a type of sugar). Glucose is a source of fuel for the body.

In people with diabetes, the body cannot regulate the amount of glucose in the blood (called blood sugar). Insulin therapy can help some people with diabetes maintain their blood sugar levels.

Information

Carbohydrates from food are broken down into glucose and other sugars. Glucose is absorbed from the digestive tract into the bloodstream. Insulin lowers blood sugar by allowing it to move from the bloodstream into muscle, fat, and other cells, where it can be stored or used as fuel. Insulin also tells the liver how much glucose to produce when you are fasting (have not had a recent meal).

People with diabetes have high blood sugar because their body does not make enough insulin or because their body does not respond to insulin properly.

- In people with type 1 diabetes the pancreas produces little to no insulin.
- In people with type 2 diabetes the fat, liver, and muscle cells do not respond correctly to insulin. This is called insulin resistance. Over time, the pancreas also stops making as much insulin.

Insulin Doses and Types

Insulin therapy replaces the insulin the body would normally make. People with type 1 diabetes must take insulin every day.

People with type 2 diabetes need to take insulin when other treatments and medicines fail to control blood sugar levels.

Insulin doses are given in two main ways:

- **Basal dose** - provides a steady amount of insulin delivered all day and night. This helps maintain blood glucose levels by controlling how much glucose the liver releases (mainly at night when the time between meals is longer).
- **Bolus dose** - provides a dose of insulin at meals to help move absorbed sugar from the blood into muscle and fat. Bolus doses can also help correct blood sugar when it gets too high. Bolus doses are also called nutritional or meal-time doses. Sometimes, a bolus dose must also be taken with large snacks.

There are several types of insulin available. Insulin types are based on the following factors:

- Onset - how quickly it starts working after injection
- Peak - time when the dose is the strongest and most effective
- Duration - total time the insulin dose stays in the bloodstream and lowers blood sugar

Below are the different types of insulin:

- **Rapid-acting or fast-acting insulin** starts working within 15 minutes, peaks in 1 hour, and lasts for 2 to 4 hours. It is taken right before or just after meals and snacks. It is often used with longer-acting insulin.
- **Regular or short-acting insulin** starts working within 30 minutes, peaks within 2 to 3 hours, and lasts 3 to 6 hours. This is taken a half-hour before meals and snacks. It is often used with longer-acting insulin.
- **Intermediate-acting insulin** starts working within 2 to 4 hours, peaks in 4 to 12 hours, and lasts 12 to 18 hours. This is taken mostly either twice a day or once at bedtime.
- **Long-acting insulin** starts to work within a few hours and works for about 24 hours, sometimes longer. It helps control glucose throughout the day. It is often combined with rapid- or short-acting insulin as needed.
- **Premixed or mixed insulin** is a combination of 2 different types of insulin. It has both a basal and bolus dose to control glucose after meals and throughout the day. It starts working in 15 to 60 minutes and lasts for 10 to 16 hours.
- **Inhaled insulin** is a rapid-acting breathable insulin powder that starts working within 15 minutes of use. It peaks 30 minutes after taking it and lasts about 3 hours. It is used just before meals.

One or more types of insulin may be used together to help control your blood sugar. You also may use insulin along with other diabetes medicines. Your health care provider will work with you to find the right combination of medicines for you.

Your provider will tell when and how often you need to take insulin. Your dosing schedule may depend on:

- Your weight
- Type of insulin you take
- How much and what you eat
- Level of physical activity
- Your blood sugar level
- Other health conditions

Your provider can calculate the insulin dose for you and teach you how to do the calculation. Your provider will also tell you how and when to check your blood sugar and time your doses during the day and night.

Ways to Take Insulin

Insulin cannot be taken by mouth because stomach acid destroys it. It is most often injected under the skin into fatty tissue. There are different insulin delivery methods available:

- **Insulin syringe** -- insulin is drawn from a vial into a syringe. Using the needle, you inject the insulin under the skin.
- **Insulin pen** -- disposable insulin pens have prefilled insulin delivered under the skin using a replaceable needle.
- **Insulin pump** -- a small machine worn on the body pumps insulin under the skin throughout the day. A small tube connects the pump to a small needle inserted into the skin.
- **Inhaler** -- a small device you use to inhale insulin powder through your mouth. It is used at the start of meals.
- **Injection port** -- a short tube is inserted into the tissue under the skin. The port containing tube is adhered to skin using adhesive tape. Fast-acting insulin is injected into the tube using a syringe or pen. This allows you to use the same injection site for 3 days before rotating to a new site.

You should talk with your provider about your preferences when deciding on an insulin delivery method.

Insulin is injected into these sites on the body:

- Abdomen
- Upper arm
- Thighs
- Hips

Your provider will teach you how to give an insulin injection or use an insulin pump or other device. It is important not to always use the same site as this can decrease how effective the insulin is for you.

Points to Remember

You need to know how to adjust the amount of insulin you are taking:

- When you exercise
- When you are sick
- When you will be eating more or less food
- When you are traveling
- Before and after surgery

When to Call the Doctor

If you are taking insulin, contact your provider if:

- You think you may need to change your insulin routine
- You have any problems taking insulin
- Your blood sugar is too high or too low and you don't understand why

Alternative Names

Diabetes - insulin

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